A Channel Adaptive Dual Siamese Network for Hyperspectral Object Tracking

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***Abstract*— Hyperspectral object tracking (HOT) aims at tracking targets using the rich spectral information from hyperspectral video (HSV). Recently, dual Siamese network (DSN) has been proposed for HOT with advanced performances, via integrating a RGB Siamese branch with a hyperspectral Siamese branch to solve small sample challenge of hyperspectral modality. However, there are still challenges of DSN that reduce its practicality: a single DSN model is difficult to process HSVs with varied channels; the spatial features extracted by the pretrained RGB branch plays a dominant role, while the hyperspectral features are not fully explored. To address the challenges, we propose a Channel AdapTive DSN, termed SiamCAT, for HOT with varied channels. Specifically, treating each frame of HSV as a grayscale image sequence varied with wavelengths, a channel adaptive (CA) module is introduced to encode the grayscale image sequence of different lengths into a uniform length, and so that SiamCAT can process HSV with varied channels. Meanwhile, a guided learning attention (GLA) module is proposed to progressively learn spectral features of the tracked target highlighted by the spatial attention of the pretrained RGB branch. Note that, to force spectral features play a leading role, instead of traditional features fusion, the spectral features extracted by the hyperspectral branch are utilized for confirming the target position. In the experiments, SiamCAT were verified by using the HOT competition dataset (i.e., 16-channel, 25-channel, and 15-channel HSVs with different wavelength ranges) and the WHU-Hi-H^3 dataset (25-channel HSVs), and achieved advanced performances.**

***Index Terms*— Channel adaptive (CA), dual Siamese network (DSN), hyperspectral object tracking (HOT), hyperspectral video (HSV).**

I. Introduction

OBJECT tracking is one of the fundamental open problems in the field of computer vision. The task is to predict the position of the target in subsequent frames of the video by artificially providing the position information of the target in the first frame. Object tracking has a wide range of applications, including autonomous driving [1], security monitoring [2], robotics [3], and sports analytics [4]. In recent years, RGB video-based object tracking has developed rapidly and achieved good performances. However, due to the limitation of only three channels of spectral information in RGB videos, RGB trackers are prone to tracking failure or drift in some challenging scenarios, such as background clutter (BC) and small target [5], [6]. With the development of spectral imaging technology, hyperspectral videos (HSVs) can now be obtained from snapshot hyperspectral cameras. Since HSV contains abundant spectral information of the target [7], [8], [9], it significantly enhances the discriminative capability and effectively separates the target from the background, even if the background is similar to the target. Therefore, HSV-based object tracking has advantages over RGB object tracking in distinguishing the target from a complex background. As a result, hyperspectral object tracking (HOT) has excellent potential in various fields, such as military reconnaissance and security surveillance [10].

In recent years, deep learning has been introduced into HOT. As is well known, deep learning is a data-driven method that requires a large amount of annotated training data to complete the model construction. However, the acquisition of HSV data is not as simple and convenient as the acquisition of RGB video data, and it requires a lot of manpower and resources. This leads to the HSV datasets being much smaller than the RGB video datasets used for object tracking, such as GOT-10K [11], LaSOT [12], and TrackingNet [13], which makes it difficult to support deep hyperspectral tracker training. Therefore, the deep hyperspectral trackers have introduced pretrained RGB tracking models to alleviate this problem. The most direct strategy is to convert hyperspectral images (HSIs) into pseudocolor images and then input them into an RGB tracking network to track the target, i.e., SiamHT [14], SEE-Net [15], and CBFF-Net [16]. Clearly, the conversion process can lead to the loss of spectral information and limits the tracking performance. In order to fully preserve the spectral information, a strategy combining hyperspectral and RGB tracking networks has emerged, specifically extracting the spectral information of the target through the hyperspectral tracking network, to improve the tracking performance of the RGB tracking network, i.e., SiamHYPER [17] and TFTN [18]. Compared to the previous strategy, this approach does not

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have a HSI conversion process, and results in a significant improvement in tracking speed. At the same time, it utilizes more spectral information and can obtain a better tracking performance.

The dual Siamese network (DSN) stands out as a representative method within the strategy of combining RGB and hyperspectral tracking networks, exhibiting excellent performance. However, is still faced with the following challenges.

1) Difficult to handle video with an arbitrary number of channels. Due to differences in the design of the filter array, hyperspectral cameras can capture HSVs that encompass varying wavelength ranges and have differing numbers of channels. Many types of snapshot hyperspectral cameras have been developed, including 16-channel hyperspectral cameras with a spectral range of 460–600 nm, 15-channel hyperspectral cameras with a spectral range of 600–850 nm, and 25-channel hyperspectral cameras with a spectral range of 665–960 nm. However, the current DSN trackers are usually designed to work effectively with HSV with a specific number of channels. It is usually necessary to retrain a new model when the number or the wavelength range of the channels changes.
2) Difficult to fully utilize hyperspectral discriminative capability. The existing DSN methods fuse spectral features from the hyperspectral branch and spatial features from the RGB branch to obtain more robust features for subsequent target localization, thereby improving the tracking performance of the RGB Siamese network. However, during the tracking process, whether in target location confirmation or bounding box regression, the tracking model exhibits a significant reliance on spatial features from RGB branch, with hyperspectral features only playing an auxiliary role. It results in the underutilization of the potential of hyperspectral features in material discrimination [19], [20], [21], which can help the tracker better identify the target. Simultaneously, the pseudocolor image input into the DSN is derived from the HSI. Therefore, the fusion of spectral and spatial features may lead to the issue of feature redundancy.

The existing DSN models are vector-based methodologies, which means that they treat the hyperspectral imagery as high-dimensional vectors [22]. More precisely, these models treat the hyperspectral imagery as an assemblage of pixel vectors, executing the tracking process within a feature space. In this context, every pixel is treated as an element in a d-dimensional feature space, where d represents the number of dimensions (channels). The existing tracking models operate within a fixed-dimensional feature space to discriminate the target and background. Nonetheless, a HSI can be regarded as a collection of organized and continuous spectral sequences within the spectral space. Hence, we have the opportunity to investigate the representation of hyperspectral imagery from a sequential standpoint. When dealing with sequential data, recurrent neural networks (RNNs) readily come to mind, which use hidden layers or memory cells to learn the states that model the underlying dynamics of the input sequence for sequential data over time. RNNs have been widely applied in many fields of HSI processing, such as HSI classification [22], HSI target detection [23], and HSI denoising [24]. Since the temporal variability of a sequential signal is similar to the spectral variability of a HSI, the same idea can be applied to hyperspectral imagery. Meanwhile, RNNs can handle sequence inputs of any length [25], suggesting that RNNs can, theoretically, process HSIs with any number of channels.

In this article, to address the above issues, we propose a channel adaptive dual Siamese network (SiamCAT) for HOT. The proposed SiamCAT tracker adopts a DSN framework, mainly consisting of a hyperspectral target aware (HTA) module and a color target localization (CTL) module. The CTL module is a pretrained RGB Siamese network. The HTA module adopts the same Siamese network architecture as the CTL module. To adapt the tracker to HSV inputs of different channel types, we design a channel adaptive (CA) module, which processes the HSI as a spectral sequence, with each channel as the input at each moment. The input sequence of the module can be of any length, so that, theoretically, the module is capable of processing HSV with any number of channels. In order to enable the tracker to more fully leverage the potential of hyperspectral features, we introduce a guided learning attention (GLA) module in the HTA module, which utilizes the spatial position information of the target learned by the CTL module to guide the HTA module in learning the corresponding hyperspectral features. It is noted that the spectral and spatial features are not fused during this process. Instead, the spatial features learned by the CTL module are employed to generate learning weights, where a larger weight represents a higher probability of the presence of the target at that specific location, as assessed by the CTL module, thereby enabling the HTA module to selectively learn pertinent features.

The main contributions of this article are summarized as follows.

1) We propose a SiamCAT for object tracking in HSV with different channel numbers, which incorporates both a hyperspectral branch and an RGB branch. To prompt spectral features to play a dominant role during tracking, the hyperspectral branch is responsible for confirming the target position without fusing spatial features from the RGB branch.
2) A CA module is proposed to adapt the tracker to HSV with different channel types. The module processes the hyperspectral imagery as spectral sequences, with each channel being a value of the sequence data. Due to the arbitrary length of the input sequence, the module can process HSV with different numbers of channels.
3) A guided learning attention (GLA) module is proposed to fully mine the target hyperspectral features without fusing spatial features. The module generates spatial attention through the spatial features extracted by the RGB branch, enabling the hyperspectral branch to selectively learn corresponding spectral features.
4) Extensive experimental results and a comprehensive analysis are presented, showing that the proposed SiamCAT tracker outperforms the advanced hyperspectral trackers.

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The remainder of this article is organized as follows. Section II introduces the related works. Section III gives the details of the proposed SiamCAT tracker. Section IV presents the experimental results and analysis. Finally, Section V concludes the article.

II. Related Works

Over the past decade, correlation filters and Siamese networks have been the two most prominent paradigms for HSV object tracking. In this section, we provide a detailed introduction to these two hyperspectral tracking paradigms.

A. Correlation Filter-Based Hyperspectral Object Tracking

In correlation filter-based tracking, a correlation filter is trained online on the region of interest by minimizing a least-squares loss function. The target is then detected in consecutive frames by convolving the trained filter via fast Fourier transform (FFT). Qian et al. [26] proposed a convolutional feature-based kernelized correlation filter method (CNHT), which uses a local HSI cube on the region of interest as a convolution kernel and performs convolution operations on the input HSI to extract the intrinsic structural features of the target. The multidimensional features are then input into the kernelized correlation filter (KCF) [27] tracker to determine the target position. Uzkent et al. [28] proposed a deep hyperspectral KCF-based tracker (DeepHKCF), which utilizes the extracted deep convolutional features replace the traditional manual features for input into the KCF tracker. Xiong et al. [19] designed the spectral-spatial histogram of multidimensional gradients (SSHMG) to describe the 3-D local spectral-spatial structure of HSIs, and combined it with the fractional abundance to represent the target material information, which is then input into the background-aware correlation filter (BACF) [29] for target tracking. Zhang et al. [30] proposed a multifeature integration-based HSV tracking (MFI-HVT) method, which combines convolutional features with histogram of oriented gradients (HOG) features as mixed features for input into the KCF tracker. Zhang et al. [31] selected three channels of hyperspectral imagery to form pseudocolor images by calculating the contrast, information entropy, and target background difference, which are then input into the BACF tracker. Although correlation filter based methods have achieved good results, the manually crafted features lack robustness and tend to exhibit tracking drift issues when confronted with complex scenes. Moreover, these models entail high computational costs, leading to a slower tracking speed.

B. Siamese Network-Based Hyperspectral Object Tracking

The Siamese network consists of two identical subbranches: one for the target template and the other for the target search region. The network takes both template and search regions as inputs and outputs the local similarity with the target for each location in the search region. According to the features used in the target modeling, the deep HOT methods can be divided into two aspects.

1) Pseudocolor Feature-Based Methods: These methods achieve target localization by extracting or mapping three channels from the hyperspectral imagery to form pseudocolor images and input them into a pretrained RGB tracker. For example, Tang et al. [32] designed a background-aware band selection (BABS) module to map the HSI to a three-channel image through matrix projection. On this basis, they replaced the BABS module with a heterogeneous encoder–decoder (HED) and added a spectral semantic representation (SSR) module to form the SiamHT tracker [14]. Wang et al. [33] selected three channels of the HSI by calculating the optimum index factor (OIF). Gao et al. [16] proposed a cross-band feature fusion network (CBFF-Net), which manually selects different band combinations of the HSI to form pseudocolor images and uses a bidirectional multiple deep feature fusion module and a cross-band group attention module to fuse the different pseudocolor image features. Li et al. [34] proposed a band grouping based Siamese HOT network (SiamBAG), which uses an optimized grouping strategy to divide different hyperspectral bands into multiple three-channel data. Li et al. [15] proposed an end-to-end deep ensemble network (SEE-Net), which introduces a spectral self-expressive module to learn the band importance, so as to form pseudocolor images.

2) Hyperspectral Feature-Based Methods : These methods achieve tracking by combining a hyperspectral tracker with an RGB tracker, wherein the entire HSI is directly fed into the hyperspectral tracker to extract hyperspectral features, thus enhancing the tracking performance of the RGB tracker. Liu et al. [35] proposed an anchor-free Siamese network (HA-Net), which introduces a spectral classification branch to obtain more discriminative features by utilizing all the channels of the HSI. Liu et al. [17] proposed a dual deep Siamese network framework (SiamHYPER) for learning a hyperspectral tracker from a pretrained RGB tracker. Su et al. [36] proposed a transformer-based three-branch Siamese network (TrTSN), where one branch is a hyperspectral branch to deal with the HSI. On this basis, Zhao et al. [37] proposed a transformer-based multimodality information transfer network (TMTNet), which adopts three-branch Siamese network as the same as the TrTSN tracker. Zhao et al. [18] proposed a transformer-based fusion tracking network (TFTN), which utilizes a self-attention mechanism to fuse the hyperspectral and pseudocolor features. In order to apply to real-time applications, Sun et al. [38] proposed a lightweight DSN (SiamOHOT) for onboard HOT, which is learned from SiamHYPER through joint spatial-spectral knowledge distillation.

The pseudocolor feature-based methods cause the loss of spectral information during the conversion process, resulting in limited tracking performance. However, the hyperspectral feature based methods can fully utilize spectral information without conversion process, which have better tracking performances and speed.

III. Channel Adaptive Dual Siamese Network

In this section, the details of the proposed SiamCAT tracker are introduced. To begin with, an introduction is provided to the general framework of SiamCAT. Subsequently, the CA module used for handling HSV with varying channel numbers

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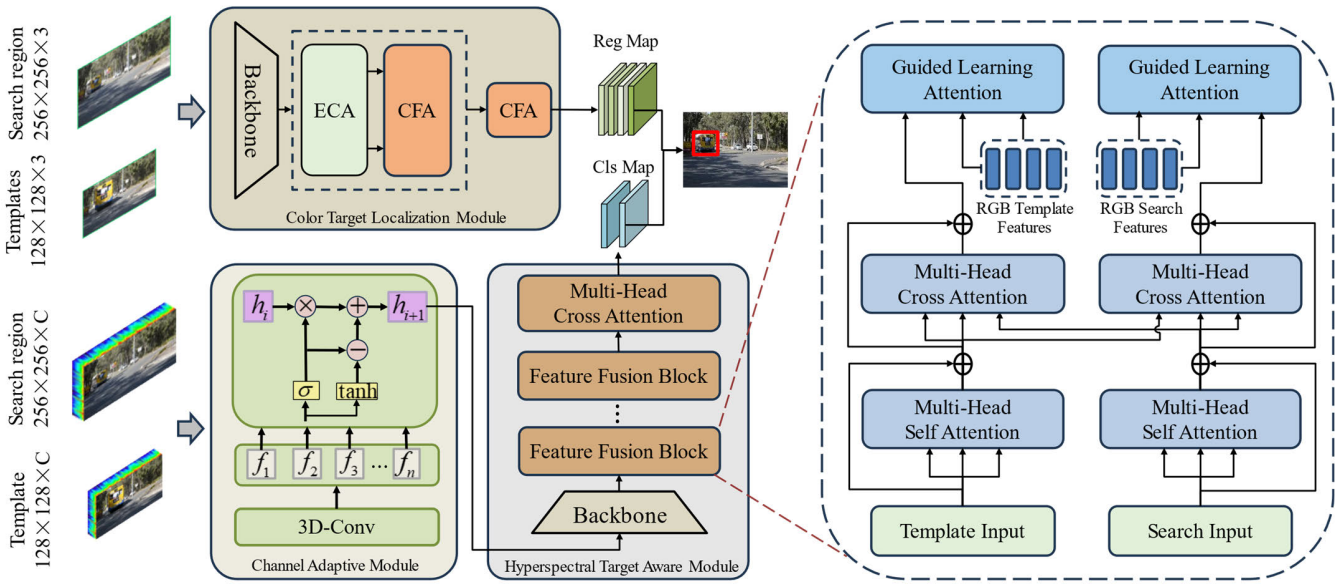


Fig. 1. Architecture of the proposed SiamCAT tracker. The channel adaptive module can adapt to input hyperspectral images of different channel types, and then the hyperspectral target aware module is used to classify candidate bounding boxes. The false-color images are input into the color target localization module for predicting the target bounding boxes. The ECA represents ego-context augment module and the CFA represents cross-feature augment module.

is presented. In addition, the GLA module is introduced to emphasize the learning of target hyperspectral features. Finally, a detailed description is given of how SiamCAT is trained and used for tracking.

A. Overall Architecture

The proposed SiamCAT tracker adopts a DSN framework, mainly consisting of a CTL module and a HTA module, as shown in Fig. 1. The CTL module is a pretrained RGB Siamese network known as TransT [39]. The structure of the HTA module is consistent with the CTL module, both of which contain three parts: feature extraction, feature fusion, and a position prediction head. The HTA module is divided into a template branch and a search branch, as in the CTL module. The difference is that the inputs of the two are different. The input of the CTL module is false-color images, while the input of the HTA module is HSIs.

1) Feature Extraction: The feature extraction part contains a template branch and a search branch. The template branch and the search branch share the weight parameters of the network, and project the input into the same feature space. The hyperspectral template patch and search patch first adjust and extract shallow spectral information through the CA module, which is then input into the backbone of the Siamese network to further extract deep spectral features. A modified ResNet-50 model [40] is used as the backbone in the proposed SiamCAT, which removes the last stage of ResNet-50 and takes the output of the fourth stage as the final output. The convolution stride of the downsampling unit of the fourth stage is also changed from 2 to 1 to obtain a higher feature resolution. In addition, the 3×3 convolution in the fourth stage is changed to dilated convolution with stride of 2 to increase the receptive field. The false-color template patch and search patch are directly input into the feature extractor of the Siamese network to extract robust spatial features.

2) Feature Fusion: The feature fusion part also contains a template branch and a search branch. The template patch and search patch features are input into the corresponding fusion branch, respectively. The attention mechanism replaces the traditional correlation operation for feature fusion, to better utilize the spectral information of the target. The feature fusion networks of the HTA module and CTL module both fuse template features and search features through two attention mechanisms: self-attention is used to enhance the feature representation, and cross-attention is used for information exchange between the template patch features and search patch features. It is worth noting that the GLA module is proposed to exchange information between the CTL module and HTA module.

3) Position Prediction: The position prediction head is divided into two tasks, namely, classification and regression. The regression head predicts the bounding box of the target, and the classification head predicts the confidence score of the candidate bounding boxes. The CTL module is responsible for the regression task, while the HTA module is responsible for the classification task during the final tracking process. It is worth noting that the HTA module operates independently of the CTL module for the classification task and does not involve the fusion of spatial features from the CTL module.

B. Channel Adaptive Module

Since HSIs can be used to utilize the difference in spectral curves between the target and background to separate them, it can better identify the target, compared to color images. Sequence data refer to the ordered collections of discrete or continuous values that convey information through a specific arrangement or pattern. They are often represented as a series of elements indexed by time or position, denoted as $\{x_{t_1}, x_{t_2}, \ldots, x_{t_n}\}$, where x_{t_i} represents the value at time step t_i, and n signifies the length of the sequence.

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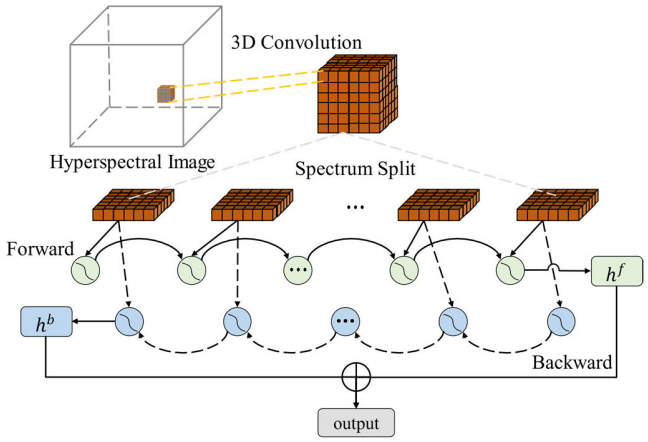


Fig. 2. Architecture of the channel adaptive module, which is composed of 3-D convolution and a quasi-bidirectional long short-term memory neural network.

Corresponding to hyperspectral imagery, x_{t_i} represents the reflection value and t_i represents the wavelength. Therefore, the spectral curve of hyperspectral imagery can be regarded as spectral sequence data. Analyzing sequence data makes it easy to think of RNNs. The ability of RNNs to model sequential dependencies and their adaptability to varying sequence lengths makes them indispensable for sequence data analysis.

In this work, we designed a CA module which constructs a modified long short-term memory (LSTM) neural network [41] to extract features in the target spectral curve, as shown in Fig. 2. The modified LSTM excludes the output gate and preserves the forget gate and input gate. At each time step, the CA module receives a channel from the processed HSV frame and performs 3-D convolution on it, then applies the corresponding activation function to obtain the forget gate $\boldsymbol{F}$ and cell state input $\boldsymbol{Z}$. According to [42], the activation function used for the cell state input $\boldsymbol{Z}$ and the forget gate $\boldsymbol{F}$ is the tanh function and sigmoid function, respectively,

$$\boldsymbol{Z} = \tanh(\boldsymbol{W}_z * \boldsymbol{I}) \tag{1}$$

$$\boldsymbol{F} = \sigma\left(\boldsymbol{W}_f * \boldsymbol{I}\right) \tag{2}$$

where $\boldsymbol{I} \in \mathbb{R}^{C\times H\times W}$ denotes the input HSI; both $\boldsymbol{W}_z$ and $\boldsymbol{W}_f \in \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}\times 3\times 3\times 3}$ are 3-D convolution kernels; $*$ denotes 3-D convolution; σ indicates a sigmoid nonlinearity.

In order to reduce model parameters, we used $\mathbf{1} - \boldsymbol{F}$ to obtain the input gate of the modified LSTM. Subsequently, the CA module selectively discards previously stored spectral information through the forget gate, and then records the spectral information for the current channel through the input gate, resulting in the current time step's cell state $\boldsymbol{h}$. At each time step, the input of the CA module consists of two parts, namely the cell state $\boldsymbol{h}$ at the previous time step and the spectral channel data at the current time step

$$\boldsymbol{h}_i = \boldsymbol{F} \odot \boldsymbol{h}_{i-1} + (\mathbf{1} - \boldsymbol{F}) \odot \mathrm{Z} \tag{3}$$

where $\boldsymbol{h}_i \in \mathbb{R}^{C_{\text{out}}\times H\times W}$ denotes the cell state feature vectors of channel i, $\boldsymbol{h}_0 = \mathbf{0}$.

Finally, the cell state at the final time step is capable of retaining spectral information from all the spectral channels. Thus, the CA module employs the final cell state as the extracted feature of the HSV frame. To enhance the extraction of spectral information and minimize spectral loss, the CA module employs a bidirectional structure, which includes both a forward pass and a backward pass. As shown in Fig. 2, the output of the CA module is calculated by adding $\boldsymbol{h}_f$ and $\boldsymbol{h}_b$ in an element-wise manner, $\boldsymbol{h}_f$ denotes the cell state features at the final channel in forward process and $\boldsymbol{h}_b$ denotes the cell state features at the final channel in backward process. Algorithm 1 describes the detailed steps of the CA module.

Algorithm 1 Channel Adaptive Module

Input: Hyperspectral video frame $\boldsymbol{I}$ with c channels
Output: Hyperspectral feature $\boldsymbol{H}$
1: Compute the cell state input $\boldsymbol{Z}$ using equation (1):
$\boldsymbol{Z} = \tanh(\boldsymbol{W}_z * \boldsymbol{I}) = (z_1, z_2, \ldots, z_c)$
2: Compute the forget gate in forward process using equation (2):
$\boldsymbol{F} = \sigma\left(\boldsymbol{W}_f * \boldsymbol{I}\right) = (f_1, f_2, \ldots, f_c)$
3: Compute the forget gate in backward process using equation (2):
$\boldsymbol{B} = \sigma(\boldsymbol{W}_b * \boldsymbol{I}) = (b_1, b_2, \ldots, b_c)$
4: Execute forward process using equation (3):
$\boldsymbol{h}_0 = 0$
For $i = 1$ to c do
$\boldsymbol{h}_i = f_i \odot \boldsymbol{h}_{i-1} + (1 - f_i) \odot z_i$
End
$\boldsymbol{h}_f = \boldsymbol{h}_c$
5: Execute backward process using equation (3):
$\boldsymbol{h}_c = 0$
For $i = c$ to 1 do
$\boldsymbol{h}_{i-1} = b_{i-1} \odot \boldsymbol{h}_i + (1 - b_{i-1}) \odot z_i$
End
$\boldsymbol{h}_b = \boldsymbol{h}_0$
6: Compute the output H:
$\boldsymbol{H} = h_f + h_b$

C. Guided Learning Attention Module

Within the DSN framework, due to the limitation in training data, relying solely on the hyperspectral branch proves challenging in extracting robust hyperspectral features of the target. It is necessary to utilize features extracted by the RGB branch as auxiliary features to aid in the learning process of the hyperspectral branch. Therefore, we designed the GLA module to implement this, as shown in Fig. 3. In the SiamHYPER algorithm, the spatial-spectral cross-attention (SSATTN) module employs a channel attention mechanism to adaptively adjust the channel responses of the false-color and HSI features, and then the adjusted channel responses of the two are weighted and fused to achieve information exchange between the two Siamese networks. Unlike SSATTN, the proposed GLA module does not involve any feature fusion process. The GLA uses the target spatial prior obtained from the CTL module to guide the HTA module to learn discriminative target hyperspectral features. The GLA module generates learning weights through the target robust spatial features, enabling the hyperspectral branch to learn the target spectral features purposefully, enhancing the model's discriminative ability.

The GLA module uses a self-attention mechanism to complete the information exchange between the two Siamese networks, which admits as inputs the RGB features $\boldsymbol{X}_{\text{rgb}}$ from the CTL module and the hyperspectral features $\boldsymbol{X}_{\text{hsi}}$ from the HTA module. The GLA module first multiplies

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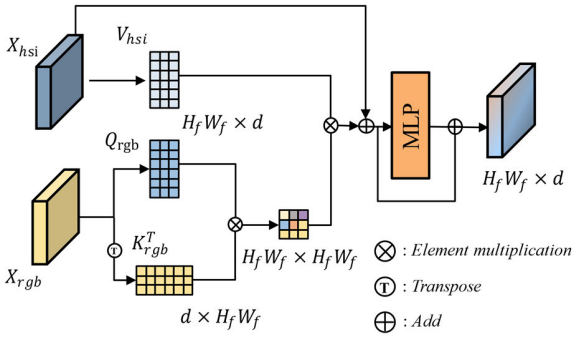


Fig. 3. Architecture of the guided learning attention module.

the transformation matrices and the input features to obtain the queries $\boldsymbol{Q}_{\text{rgb}} \in \mathbb{R}^{H_f W_f \times d}$, the keys $\boldsymbol{K}_{\text{rgb}} \in \mathbb{R}^{H_f W_f \times d}$ and the values $\boldsymbol{V}_{\text{hsi}} \in \mathbb{R}^{H_f W_f \times d}$ of the self-attention mechanism

$$\boldsymbol{Q}_{\text{rgb}} = \boldsymbol{X}_{\text{rgb}} \boldsymbol{W}^q_{\text{rgb}} \tag{4}$$

$$\boldsymbol{K}_{\text{rgb}} = \boldsymbol{X}_{\text{rgb}} \boldsymbol{W}^k_{\text{rgb}} \tag{5}$$

$$\boldsymbol{V}_{\text{hsi}} = \boldsymbol{X}_{\text{hsi}} \boldsymbol{W}^v_{\text{hsi}} \tag{6}$$

where $\boldsymbol{X}_{\text{rgb}} \in \mathbb{R}^{H_f W_f \times d}$ and $\boldsymbol{X}_{\text{hsi}} \in \mathbb{R}^{H_f W_f \times d}$ denote the RGB branch features and hyperspectral branch features, respectively; $\boldsymbol{W}^q_{\text{rgb}} \in \mathbb{R}^{d \times d}$, $\boldsymbol{W}^k_{\text{rgb}} \in \mathbb{R}^{d \times d}$, and $\boldsymbol{W}^v_{\text{hsi}} \in \mathbb{R}^{d \times d}$ denote the corresponding transformation matrices; and d denotes the dimension of the features. $H_f = (H/8)$, $W_f = (W/8)$.

Then, it obtains the learned attention weights through the element multiplication of $\boldsymbol{Q}_{\text{rgb}}$ and $\boldsymbol{K}^T_{\text{rgb}} \in \mathbb{R}^{d \times H_f W_f}$. The learned attention weights indicate that the RGB branch is more likely to identify which areas are the target. Finally, the GLA module multiplies the learned attention weights and the $\boldsymbol{V}_{\text{hsi}}$ to enable the hyperspectral branch to learn the target spectral features purposefully. The process can be denoted as:

$$\text{GLA}(\boldsymbol{X}_r, \boldsymbol{X}_h) = \text{softmax}\left[\frac{\boldsymbol{Q}_{\text{rgb}} \boldsymbol{K}^T_{\text{rgb}}}{\sqrt{d}}\right] \boldsymbol{V}_{\text{hsi}}. \tag{7}$$

D. Tracking With SiamCAT

1) Training: The RGB tracker TransT [39] used in SiamCAT is an off-the-shelf model provided by the corresponding authors, while the CA module and the HTA module are trained from scratch, without any pretrained model. During the training process, we load and freeze the parameters of the TransT model. The TransT model was trained on the COCO [43], TrackingNet [13], LaSOT [12], and GOT10k [11] datasets, which the backbone parameters are initialized with ImageNet-pretrained ResNet-50 and other parameters are initialized with Xavier init. The binary cross-entropy loss is implemented as the loss function for the classification, which is defined as:

$$L_{\text{cls}} = -\sum_j \left[y_j \log p_j + (1 - y_j) \log(1 - p_j) \right] \tag{8}$$

where y_j denotes the ground-truth label of the jth sample, $y_j = 1$ denotes the foreground and $y_j = 0$ denotes the background; and p_j denotes the probability of belonging to the foreground, as predicted by the learned model.

For the bounding box regression, a linear combination of l_1-norm loss $L_1(.,.)$ and the generalized intersection over union (IoU) loss [44] $L_{\text{GIoU}}(.,.)$ is employed as the loss function, which can be formulated as:

$$L_{\text{reg}} = \sum_j 1_{[y_i=1]} \left[\lambda_G L_{\text{GIoU}}(b_j, \hat{b}) + \lambda_1 L_1(b_j, \hat{b}) \right] \tag{9}$$

where $y_j = 1$ denotes the positive sample, b_j denotes the jth predicted bounding box, and $\hat{b}$ denotes the normalized ground-truth bounding box. $\lambda_G = 2$ and $\lambda_1 = 5$ are the regularization parameters in the experiments [39], [45], [46].

2) Tracking: The template is obtained in the first frame of the video, and the search images are taken from the following frames. The target location is estimated with the classification response maps output by the classification branch of the HTA module. The classification response maps contain two channels, which represent the probabilities of the samples being target or background. The bounding box is calculated with the regression response maps output from the regression branch of the CTL module. The regression response maps contain four channels, which represent the normalized coordinates. A penalty matrix based on the Hamming window is implemented on the classification response maps, and the weight value of the penalty matrix is set to 0.46.

IV. Experiments

A. Datasets and Evaluation Metrics

1) HOT Competition Dataset: The HOT competition dataset includes 16-channel, 15-channel, and 25-channel HSVs with wavelength ranges of 460–600, 600–850, and 665–960 nm, respectively. The dataset is made up of 95 sets of videos designated for training, along with an additional 77 sets of videos designated for validation. Each set contains HSVs and false-color videos converted from HSVs. The dataset is available at the competition website.[1]

2) WHU-Hi-H^3 Dataset: The WHU-Hi-H^3 dataset [47] contains 69 sets of annotated HSVs with a total of 25 channels, all of which were used for the validation. Each set only contains HSVs. Moreover, the dataset is annotated by nine attributes corresponding to different challenging factors: fast motion (FM), BC, illumination variation (IV), occlusion (OCC), motion blur (MB), low resolution (LR), scale variation (SV), out-of-view (OV), and deformation (DEF).

3) Evaluation Metrics: The precision plot, success plot, distance precision (DP) @ 20 pixels, and the area under the curve (AUC) are used here to evaluate the performance of the different trackers. A precision plot shows the percentage of frames whose estimated locations lie within a given threshold distance to the ground-truth centers. The DP @ 20 pixels score is reported at a threshold of 20 pixels. A success plot shows the percentages of successful frames whose overlap ratio between the predicted bounding box and ground truth is larger than a certain threshold, which varies from 0 to 1. The AUC score is calculated on each success plot.

B. Implementation Details

In the experiments, the proposed method was implemented in Python with the PyTorch framework. We used two NVIDIA

[1] https://www.hsitracking.com

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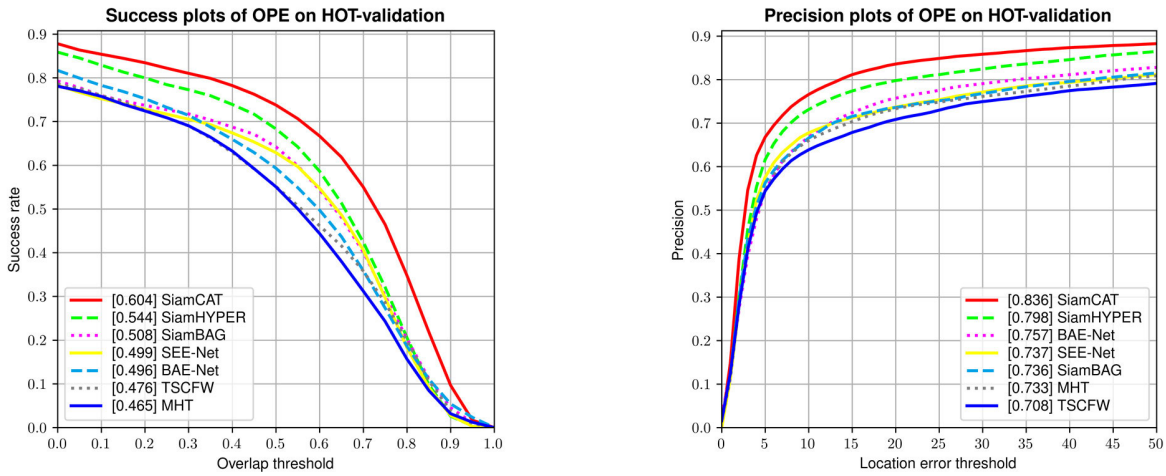


Fig. 4. Comparison with hyperspectral trackers on three channel types of hyperspectral video from the HOT competition dataset. (a) Success plot. (b) Precision plot.

TABLE I
TRACKING RESULTS OF ALL THE HYPERSPECTRAL TRACKERS ON THE HOT COMPETITION DATASET

Videos	Evaluation indicator	MHT	TSCFW	BAE-Net	SiamBAG	SiamHYPER	SEE-Net	SiamCAT (proposed)
16 channels	AUC	0.505	0.533	0.558	0.554	0.590	0.561	**0.593**
	DP	0.765	0.762	0.802	0.779	0.815	0.784	**0.822**
15 channels	AUC	0.384	0.316	0.397	0.374	0.430	0.363	**0.480**
	DP	0.516	0.416	0.508	0.517	0.579	0.496	**0.665**
25 channels	AUC	0.435	0.450	0.438	0.485	0.515	0.455	**0.665**
	DP	0.763	0.734	0.779	0.751	0.852	0.753	**0.920**

RTX 3090 graphics processing units (GPUs) for the training. For the proposed method, the AdamW optimizer was used to train the model. The learning rate for the network parameters was set to 1×10^{-5}, the batch size was set to 16, and the weight decay was set to 1×10^{-4}, for a total of 30 training epochs. The input sizes of the template branch and search branch of the CTL module were $128 \times 128 \times 3$ and $256 \times 256 \times 3$, respectively. The input sizes of the template branch and search branch of the HTA module were $128 \times 128 \times C$ and $256 \times 256 \times C$, respectively, where C represents the number of channels of the input HSV. The proposed method was trained using the training set of the HOT competition dataset exclusively, while the validation was performed using the validation set of the HOT competition dataset and the WHU-Hi-H^3 dataset. During training, the three channel types of HSV were used as the input.

C. Comparison With Hyperspectral Trackers

In this experiment, the SiamCAT tracker was evaluated with several open-source hyperspectral trackers: MHT [19], TSCFW [48], BAE-Net [49], SiamBAG [34], SiamHYPER [17], and SEE-Net [15]. All the compared hyperspectral trackers were specifically designed for object tracking with 16-channel HSVs. Except for MHT and TSCFW, we retrained each of the compared methods on the three channel types of HSV. The training parameters for all the models were derived from the parameters provided by the authors. To ensure a fair comparison as much as possible, we maintained a consistent training data volume by employing interpolation to supplement the data. For example, when training the 16-channel HSV, in addition to the original 16-channel video data, we also performed channel interpolation on the 15-channel and 25-channel video to convert them into 16-channel video to augment the dataset. The MHT and TSCFW are online learning methods based on correlation filter, which are no need for retraining. The success plot and precision plot of the proposed SiamCAT tracker and the other hyperspectral trackers on the all HSVs are shown in Fig. 4. It can be observed that the proposed SiamCAT tracker achieves the best performance in both the AUC score and DP @ 20 pixels score, which confirms the effectiveness of SiamCAT in HOT. Table I shows the tracking results of the proposed SiamCAT and comparison HOT methods on each channel type of HSV. It can be observed that the tracking performance of the comparison HOT methods is not ideal compared with 16-channel HSVs when deals with 15-channel or 25-channel HSVs. The main reason maybe that the comparison HOT methods are designed for object tracking with 16-channel HSVs. It is difficult to

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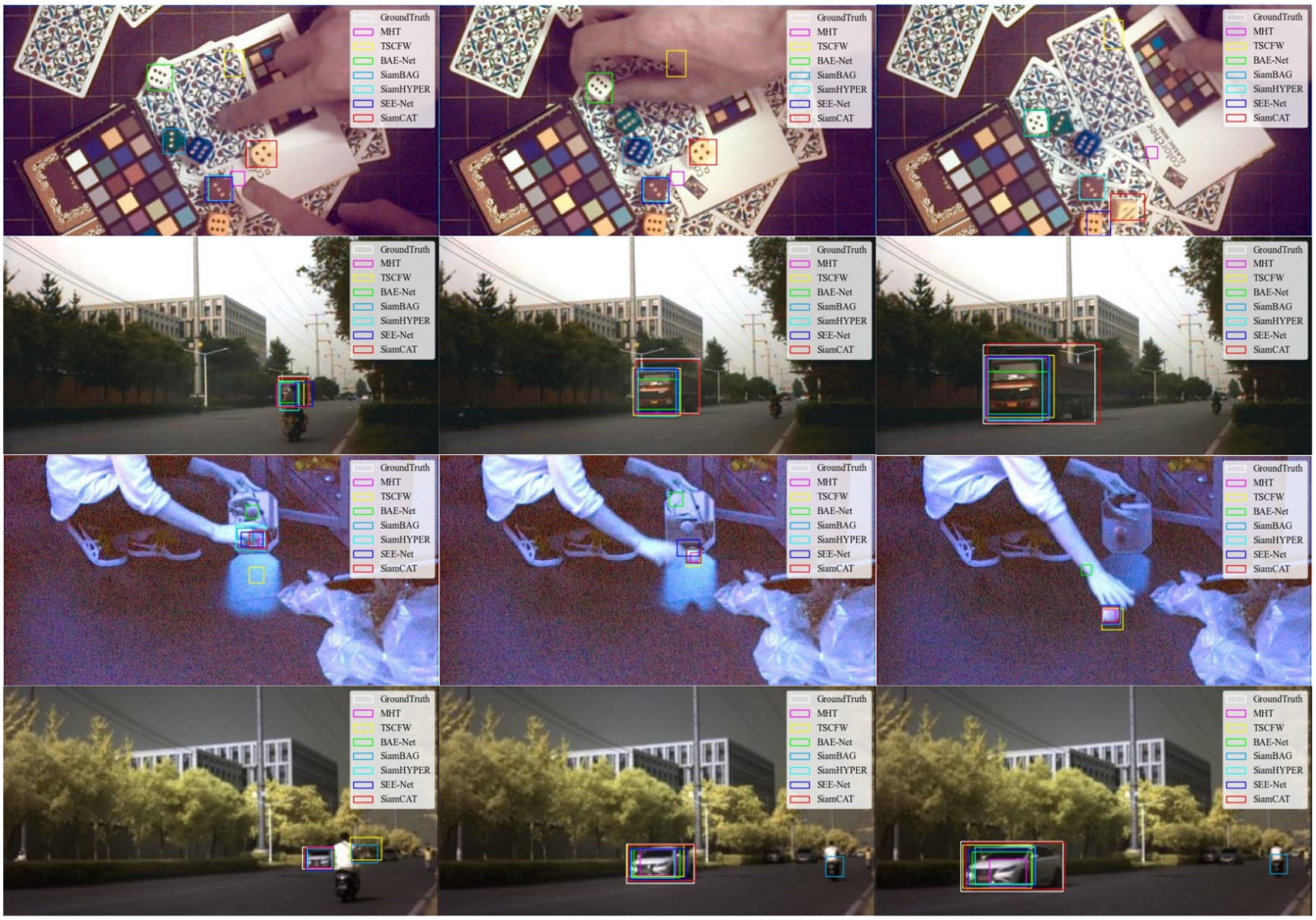


Fig. 5. Qualitative comparison of the proposed tracker on four video sequences from the HOT competition dataset (VIS-dice2, VIS-trucker, RedNIR-ball&mirror9, and NIR-car11) with MHT, TSCFW, BAE-Net, SiamBAG, SiamHYPER, and SEE-Net. The proposed approach performs favorably against these trackers.

ensure desirable tracking performance when transferring the compared methods to 15-channel or 25-channel HSVs without any adjustments. Because it remains challenging to maintain uniform training parameters across each model for different inputs. If ideal tracking performance is desired, it necessitates appropriate adjustments to the model architecture or training parameters during retraining, which is resource-consuming and laborious. It is the significant challenge faced by the current tracking methods and the issue that we aim to address. The proposed SiamCAT has universality which can handle HSVs with different channel numbers without retraining. Meanwhile, on 16-channel HSVs, as a universal method, the proposed SiamCAT can also achieve the best tracking performance compared the other hyperspectral trackers designed for the 16-channel HSVs. It indicates that the proposed GLA module directs the model's focus toward the acquisition of the target's hyperspectral features, enhancing the model's discriminative ability and enabling more accurate target localization.

To more comprehensively show the advantages of the proposed SiamCAT tracker, we further present the visual results of all the hyperspectral trackers in Fig. 5. As shown in Fig. 5, the proposed SiamCAT tracker can better handle the challenging scenarios such as scale variation (VIS-trucker, NIR-car11), BC (VIS-dice2), and low resolution (VIS-dice2, RedNIR-ball&mirror9). In the VIS-dice2 sequence, it can be observed that all comparative methods lose the target as they fail to adequately exploit the target's hyperspectral features, leading to misidentification. Conversely, the proposed method demonstrates a more effective utilization of hyperspectral features for accurate target identification. Furthermore, it can be observed that in the VIS-trucker and NIR-car sequences, the target occurs a change in scale transitioning from a distant to a close proximity. The comparative methods maintain a constant scale for tracking, whereas the proposed method demonstrates the ability to adapt to changes in the target scale. This experiment further shows the effectiveness of the SiamCAT tracker in HOT.

To verify the generalization of the proposed SiamCAT tracker, we conducted experiments on the WHU-Hi-H^3 dataset. Specifically, we directly tested the deep learning models trained on the HOT competition dataset with any fine-tuning operations. The quantitative evaluation results are listed in Table II. It can be seen that the SiamCAT tracker shows the best tracking performance, which verifies that the proposed model remains effective when transferred to other datasets.

D. Comparison With the State-of-the-Art RGB Trackers

In this experiment, the SiamCAT tracker was evaluated with several state-of-the-art RGB trackers, i.e., DropTrack [50], OSTrack [51], Mixformer [45], TransT [39], Stark [52],

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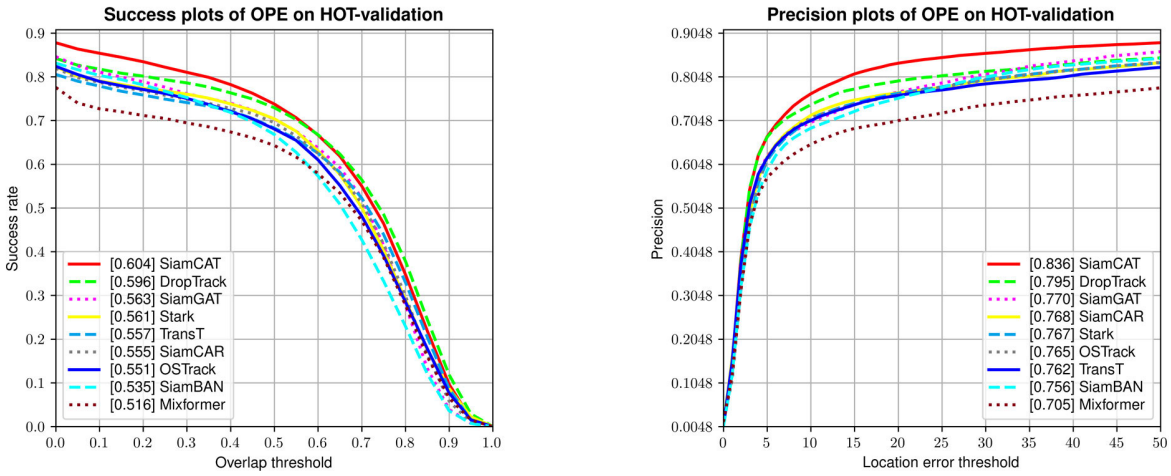


Fig. 6. Comparison with RGB trackers on false-color videos from the HOT competition dataset. (a) Success plot. (b) Precision plot.

TABLE II
QUANTITATIVE RESULTS ON THE WHU-HI-H^3 DATASET

Method	AUC	DP
MHT	0.558	0.712
TSCFW	0.568	0.723
BAE-Net	0.600	0.816
SiamBAG	0.560	0.725
SiamHYPER	0.652	0.868
SEE-Net	0.611	0.842
SiamCAT (proposed)	**0.705**	**0.890**

TABLE III
COMPARISON IN TERMS OF TRACKING SPEED OF THE HYPERSPECTRAL TRACKERS

Method	MHT	TSCFW	BAE-Net	Siam BAG	Siam HYPER	SEE-Net	Siam CAT
FPS	0.5	4.1	0.8	14.1	29.8	16.8	23.9
Device	CPU		GPU				

SiamGAT [53], SiamBAN [54], and SiamCAR [55]. All the RGB trackers used pretrained models provided by the corresponding authors. For the HOT competition dataset, all the compared RGB trackers were run on the false-color videos. The success plot and precision plot of the SiamCAT tracker on the HSV and the RGB trackers on the false-color video are shown in Fig. 6. According to the experimental results shown in Fig. 6, the SiamCAT tracker can outperform all the compared trackers. Although the AUC score of DropTrack is similar to that of the proposed method, the proposed method significantly outperforms DropTrack in terms of the DP @ 20 pixels score. This indicates that leveraging the rich spectral information in HSV can enhance the model's discriminative ability for the target.

E. Running Time Comparison

We report the tracking speed in terms of frames per second (FPS) for all the hyperspectral trackers. The MHT method was run on a Windows machine with the configuration of an AMD Ryzen 5 3600 CPU @ 3.59 GHz and 16 GB RAM. The deep hyperspectral trackers were run on a machine equipped with an Intel(R) Xeon(R) Gold 5218R CPU @ 2.10 GHz and an NVIDIA RTX 3090 GPU. Table III lists the tracking speed of all the hyperspectral trackers in the HOT competition dataset, where it can be observed that the proposed SiamCAT tracker is second only to SiamHYPER, achieving a speed of 23.9 FPS. The shooting frame rate of the datasets is 25 FPS, thereby the SiamCAT basically meets the real-time requirement. The main reason for this is that the SiamCAT tracker incorporates the self-attention mechanism for fusing the template and search features, which consumes a significant amount of time.

F. Ablation Study

We studied the impact of each module on the tracking performance. All the experiments were conducted on the HOT competition dataset.

1) Effectiveness of the Proposed CA Module: In this experiment, we replaced the CA module with other approaches of processing sequence data, i.e., averaging and the gated recurrent unit (GRU) [56]. The averaging operation refers to a straightforward spectral channel-wise averaging operation conducted on the HSI after it has undergone 3-D convolution. The GRU refers to the approach of replacing the modified LSTM with a modified GRU in the CA module. As with the modified LSTM, the modified GRU discards the previous time step's hidden state and directly uses the current time step's input to construct the reset gate R and update gate U. The update expression can be formulated as: $h_i = (1 - U) \odot h_{i-1} + U \odot (Z + R \odot h_{i-1})$. As with the CA module, the GRU also employs a bidirectional structure for feature extraction. Table IV summarizes the AUC score and DP @ 20 pixels score for the different settings. Compared to the GRU-like approach and the averaging approach, the LSTM-like approach results in an improvement of 2.1% and 3.0%, respectively. The experimental results confirm the effectiveness of the CA module.

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TABLE IV
IMPACT OF DIFFERENT SEQUENCE PROCESSING APPROACHES

Method	AUC	DP
Averaging	0.568	0.779
GRU	0.583	0.806
CA module (proposed)	**0.604**	**0.836**

TABLE V
ABLATION STUDY FOR THE MODULES OF SIAMCAT

Module	AUC	ΔAUC	DP	ΔDP
CTL	0.557	-	0.762	-
CTL+HTA	0.572	+1.5%	0.788	+2.6%
CTL+HTA+GLA	0.604	+3.2%	0.836	+4.8%

2) Effectiveness of the Modules in SiamCAT: The baseline in this experiment was the CTL module, which corresponds to the RGB tracker TransT. As shown in Table V, the results for the baseline were obtained by directly inferring with the pretrained model provided by the authors on false-color video. On this basis, we added the HTA module, and it can be observed that the AUC score increases by 1.5% and the DP @ 20 pixels score improves by 2.6%. This indicates that the abundant spectral information from HSV enables the model to more accurately distinguish the target. We have added the CA module by default to handle HSV with different channel numbers. Subsequently, the AUC score increases by an additional 3.2% and the DP @ 20 pixels score increases by an additional 4.8% after adding the GLA module. These results validate the effectiveness of the GLA module, which directs the HTA module to focus more on learning the spectral features of potential target regions, thereby enhancing the model's discriminative capability.

V. CONCLUSION

In this article, we have proposed a SiamCAT for HOT. Inspired by the capacity of RNNs to handle sequences of variable lengths, the proposed CA module employs a modified LSTM network to extract spectral features of the hyperspectral imagery with varied channels. By treating the hyperspectral imagery frame in the video as a spectral sequence data, the CA module enables the SiamCAT tracker to effectively handle HSVs with arbitrary channels. To better leverage the role of spectral information for target identification, the proposed guided learning attention module utilizes the target robust spatial features from the RGB branch to generate learning weights, which facilitates the hyperspectral branch to acquire the target spectral features. The experimental results on HOTC and WHU-Hi-H^3 datasets verify the effectiveness of the proposed SiamCAT tracker, which shows a better tracking performance than the existing hyperspectral trackers.

Nevertheless, the proposed SiamCAT exhibits certain limitations. First, it is difficult to handle HSVs with a very large number of channels, as the LSTM network may exhibit gradient vanishing phenomenon. Second, due to the introduction of the attention mechanism for feature fusion, the computational complexity has increased, resulting in the tracking speed failing slightly short of meeting real-time requirements. For our future work, we will explore how to integrate the transformer [57] into the processing of inputs from HSV with varied channel numbers. Compared to RNNs, the transformer can handle longer spectral sequences, which does not experience the problem of gradient vanishing. However, this would lead to an increase in computational complexity, potentially resulting in a reduced tracking speed, which may be addressed through formulating a lightweight network architecture.

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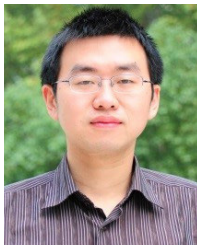
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